Article

EvoRiri: Risk-Aware Genetic Optimization of a Safety-First, Empathetic Bangla Mental-Health Assistant

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Abstract: Large language models are increasingly deployed in emotionally sensitive applications such as mental-health support, where optimization must jointly consider empathy, safety, and conversational structure. We present a genetic-algorithm (GA) framework for optimizing the response-generation policy of *Riri*, a Bangla/Banglish/English mental-health assistant. Instead of updating model weights, the GA searches over an interpretable genome that parametrizes response structure and style within a risk-aware generator. A non-differentiable fitness function combines automatic scores for empathy, safety, and structure, subject to hard safety constraints and smooth penalty terms. Experiments on a corpus of 2297 de-identified prompts with suicide-risk labels show that the GA converges rapidly to high-fitness policies, improving structural completeness by more than a factor of four while preserving high safety and slightly increasing empathy. Performance is effectively stable beyond approximately 300–600 training prompts, indicating that moderate datasets suffice to provide a stable optimization signal. Trace analysis reveals typical evolutionary failure modes including question-only structural collapse and empathy marker stuffing and demonstrates how revised scoring and progressive structure enforcement steer search toward behaviorally meaningful responses. The results highlight both the promise and the pitfalls of evolutionary optimization for value-laden conversational AI.

Keywords: evolutionary algorithms; genetic algorithms; mental-health chatbots; large language models; empathy; safety; multi-objective optimization; conversational agents; Bangla; risk-aware AI

1. Introduction

Large language models (LLMs) are increasingly used as conversational agents in domains such as mental-health support, coaching, and counseling [1,2]. Deploying these systems safely requires more than grammatical correctness or task accuracy; responses must be empathetic, structurally helpful, and aligned with risk-aware safety principles [3–5].

Conventional fine-tuning and reinforcement learning from human feedback (RLHF) are powerful but expensive and tightly coupled to model weights. In contrast, evolutionary algorithms (EAs) can optimize high-level, interpretable policies using arbitrary non-differentiable objectives [6,7]. In the context of conversational agents, this makes them attractive for optimizing template-based generators or routing strategies while keeping the underlying LLM fixed [8,9].

Unlike prior evolutionary prompt optimization approaches that focus primarily on reasoning accuracy, feature activation, or prompt engineering, the proposed framework focuses on optimizing emotionally sensitive conversational behavior under explicit safety constraints. The system introduces risk-aware escalation, progressive structure enforcement, and trace-level behavioral analysis for a multilingual mental-health assistant. To the best of our knowledge, this is among the first studies applying interpretable genetic optimization to empathy–safety–structure balancing in Bangla mental-health conversational systems.

In this article we introduce an evolutionary optimization framework for *Riri*, a mental-health-oriented assistant designed for Bangla-speaking users. The system wraps the LLM

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in a structured generator that controls when to include empathy, grounding, action suggestions, and reflective questions. A genetic algorithm (GA) evolves a genome of continuous and discrete parameters that govern this generator. The optimization objective aggregates scores for empathy, safety, and structure under explicit constraints for suicide and self-harm risk.

The main contributions of this work are:

- A risk-aware GA formulation for mental-health response generation, including an interpretable genome, progressively structured generator, and constrained fitness.
- An empirical study of GA convergence, data efficiency, and population-size sensitivity on a corpus of 2297 de-identified mental-health prompts with risk labels.
- A trace-based analysis of evolutionary failure modes (structural collapse, empathy overfitting) and demonstration of how revised scoring and hard constraints yield behaviorally meaningful responses.

The novelty of this work lies not only in applying a GA to response generation, but in using evolutionary search to optimize safety-critical conversational behavior under interpretable constraints. The framework combines risk-aware escalation, progressive structure enforcement, and trace-based failure analysis. This enables the study to examine how evolutionary systems behave when optimizing value-laden objectives such as empathy, safety, and structured support.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the dataset, genome, generator, and fitness function. Section 4 presents experimental results. Section 5 discusses implications and limitations, and Section 6 concludes.

1.1. Research Hypotheses

This study evaluates the following hypotheses:

H1: Genetic Algorithm optimization improves overall response quality compared with the baseline response generator.

H2: Progressive structure enforcement improves structural completeness without substantially reducing safety.

H3: Moderate dataset sizes provide a sufficiently stable fitness signal for GA-based response optimization.

H4: Larger GA populations improve optimization stability up to an intermediate range, after which performance gains diminish.

H5: Hard safety constraints reduce unsafe or inappropriate optimization shortcuts in high-risk mental-health contexts.

2. Related Work

2.1. Evolutionary Algorithms for LLM and Prompt Optimization

Recent research has increasingly explored the integration of Evolutionary Algorithms (EAs) with Large Language Models (LLMs) for optimization, reasoning, and controllable generation tasks. A growing body of work suggests that evolutionary search provides a promising alternative to gradient-based alignment methods, particularly in black-box API settings where direct model fine-tuning is impractical.

The survey "When Large Language Models Meet Evolutionary Algorithms" [7] establishes a conceptual bridge between token-sequence generation in LLMs and population-based search in evolutionary systems. The authors argue that prompts, policies, and language structures can be treated as evolvable individuals within an optimization landscape. Their framework motivates the use of mutation, crossover, and fitness shaping in language-space optimization. The present work follows this paradigm by treating conversational behavior policies as evolvable genomes composed of interpretable parameters such as empathy weight, safety weight, structure weight, escalation thresholds, and template configurations.

Similarly, Promptbreeder [10] proposes a self-referential prompt evolution framework in which mutation operators are themselves evolved alongside the prompts. Promptbreeder demonstrates that genetic search can discover diverse, high-performing prompts across reasoning tasks without manual engineering. While Promptbreeder targets task accuracy, the proposed framework extends the evolutionary optimization paradigm toward safety-critical conversational behavior.

GAAPO [11] further demonstrates the effectiveness of Genetic Algorithms for automatic prompt optimization in LLM systems. Their work explores mutation, crossover, selection strategies, and population hyperparameter trade-offs across benchmark reasoning tasks such as ETHOS, MMLU-Pro, and GPQA. Similar to GAAPO, the present study investigates population-size and dataset-size sensitivity in evolutionary optimization. However, unlike GAAPO, the proposed framework focuses on multi-objective optimization for mental-health dialogue systems, balancing empathy, safety, structural completeness, and conversational naturalness.

GenDLN [9] applies evolutionary optimization to stacked LLM reasoning pipelines using non-differentiable objectives. The work demonstrates that GA-based optimization can effectively operate over black-box API systems without gradient access. However, GenDLN primarily targets logical reasoning and classification performance, whereas the present work investigates emotionally sensitive conversational optimization under safety constraints.

Another closely related direction is Multimodal LLM-assisted Evolutionary Search (MLES) [8], which combines LLM reasoning with evolutionary optimization to generate interpretable programmatic control policies. MLES emphasizes behavioral evidence and transparent policy evolution rather than relying solely on scalar reward signals.

A recent systematic survey on LLMs for Evolutionary Optimization [6] categorizes hybrid EA-LLM systems into low-level and high-level optimization frameworks. The proposed framework belongs primarily to the low-level LLM-assisted optimization category, where evolutionary search operates directly over interpretable conversational policies.

Overall, existing EA+LLM research primarily focuses on reasoning accuracy, feature activation, program synthesis, or prompt engineering. The proposed work contributes to this emerging direction by introducing a risk-aware, multi-objective GA framework for mental-health conversational optimization.

2.2. Mental-Health Conversational Agents

Mental-health chatbots and conversational agents have received significant attention in recent years due to their potential to provide scalable psychological support. Systematic reviews [1,2] suggest that generative conversational agents can improve emotional accessibility and user engagement, particularly among populations with limited access to professional mental-health care.

Recent work on empathic conversational agent design [3] highlights the importance of empathy, conversational structure, and safety mechanisms in determining user trust and therapeutic effectiveness. These findings directly motivate the three-dimensional optimization objective used in the proposed framework: empathy, safety, and structural support.

Emotionally intelligent chatbot systems have also raised important ethical and safety concerns [12]. Prior studies emphasize that emotionally persuasive conversational systems can unintentionally produce unsafe or misleading outputs if optimization objectives are poorly specified.

Other mental-health chatbot systems have relied primarily on sentiment analysis and rule-based support generation [13]. In contrast, the proposed framework employs evolutionary optimization to dynamically shape conversational behavior based on fitness feedback.

The American Psychological Association (APA) advisory on generative AI for mental health [4] further stresses the necessity of explicit safety constraints and escalation policies in AI mental-health systems.

Empirical studies on the safety of generative AI chatbots in mental-health contexts [5] also demonstrate that conversational agents may exhibit inconsistent crisis behavior when not explicitly constrained.

Human-AI collaborative mental-health systems [14] have additionally shown that empathy and conversational structure significantly influence perceived support quality.

2.3. Empathy Modeling and Conversational Behavior Optimization

Empathy modeling has become an increasingly important area within conversational AI research. Surveys on artificial empathy classification [15] discuss various methods for evaluating empathetic language using linguistic markers, neural scoring systems, and psychological frameworks. The present work adopts a lightweight rubric-based evaluation approach due to its interpretability and compatibility with black-box optimization settings.

Neural empathetic conversational systems [16] have demonstrated strong performance in emotionally aware dialogue generation, but these methods often require large-scale supervised training and expensive fine-tuning pipelines. In contrast, the proposed framework explores whether evolutionary search over interpretable conversational policies can improve empathetic behavior without retraining the underlying LLM.

Research on conversational trust and chatbot evolution [17] further suggests that users respond more positively to systems exhibiting balanced empathy, clear structure, and contextual appropriateness.

2.4. Research Gap and Positioning

Existing studies on evolutionary prompt optimization primarily focus on improving reasoning performance, feature activation, or general task accuracy. Similarly, prior mental-health conversational systems often emphasize either empathy or safety independently, with limited attention to structural balance and risk-aware escalation. There remains a gap in developing interpretable optimization frameworks that jointly balance empathy, safety, structure, and adaptive risk-sensitive behavior in multilingual mental-health support systems.

Table 1. Comparison of the proposed framework with related approaches.

Approach	**GA/EA**	**Empathy**	**Safety**	**Structure**	**Risk-aware**
Rule-based mental-health chatbots	No	Partial	Partial	Partial	Partial
LLM-based mental-health assistants	No	Yes	Partial	Partial	Partial
RLHF-based alignment methods	No	Yes	Yes	Limited	Partial
GA-based prompt optimization (e.g., Promptbreeder, GAAPO)	Yes	Limited	Limited	Limited	No
Proposed framework	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

As shown in Table 1, existing approaches typically optimize either conversational quality or alignment objectives independently. In contrast, the proposed framework jointly optimizes empathy, safety, structural coherence, and risk-aware escalation within a unified evolutionary optimization framework.

3. Methodology

3.1. Problem Formulation

Let X denote the space of user prompts and $R = \{\mathsf{low}, \mathsf{mid}, \mathsf{high}\}$ the set of suicide/self-harm risk labels. A response-generation policy is a function

$$\pi_\theta : X \times R \to Y, \tag{1}$$

mapping a prompt $x \in X$ with risk $r \in R$ to a textual response $y \in Y$. Instead of optimizing LLM parameters directly, we parameterize π_θ via a higher-level generator G controlled by a genome $g \in \mathcal{G}$:

$$y = G(x, r; g). \tag{2}$$

The design goal is to find a genome $g^\star$ that maximizes a scalar fitness function $F(g)$ combining empathy, safety, and structure:

$$g^\star = \arg\max_{g \in \mathcal{G}} F(g) \quad \text{subject to safety constraints.} \tag{3}$$

We approximate (3) using a GA operating on a finite set of prompts.

3.2. Dataset

We use a dataset of $N = 2297$ de-identified prompts collected from a mental-health screening and support application in Bangladesh. Each example $i \in \{1, \ldots, N\}$ consists of:

$$(x_i, r_i, t_i), \tag{4}$$

where x_i is the user message in Bangla, Banglish, or English; $r_i \in R$ is the risk label (low, mid, high); and t_i is a high-level theme description (e.g., economic stress, relationship issues, suicidal ideation) used only for analysis [1].

For some experiments we consider subsets of size $N_s \in \{100, 300, 600, 1200, 2297\}$ to study data efficiency. Each subset is split into an optimization set and a held-out validation set.

3.3. Genome Representation

An individual genome g is a vector of continuous and discrete genes:

$$g = \left(w_e,\, w_s,\, w_{sa},\, \theta_{\text{mid}},\, \theta_{\text{high}},\, \lambda,\, d\right), \tag{5}$$

where:

- $w_e \in [0,1]$: empathy weight controlling the number and intensity of empathy lines.
- $w_s \in [0,1]$: structure weight controlling inclusion of grounding, action, and questions.
- $w_{sa} \in [0,1]$: safety weight governing conservative phrasing and crisis emphasis.
- $\theta_{\text{mid}}, \theta_{\text{high}} \in [0,1]$: risk thresholds for triggering crisis-style responses.
- $\lambda \in [0,1]$: length penalty coefficient.
- d: discrete template selector, indexing a small set of response templates.

To establish a meaningful performance reference, we define three baseline conditions. The zero-shot baseline (B1) instantiates the generator with minimum genome weights ($w_e = w_s = w_c = 0.10$, $\lambda = 0$), a short memory window of 256 tokens, and maximally conservative escalation thresholds ($\theta_{\text{mid}} = 0.70$, $\theta_{\text{high}} = 0.95$), representing response generation with no parameter optimisation. The fixed seed baseline (B2) uses the balanced initialisation genome g_{bal} without evolutionary search, isolating the contribution of the search process itself. The random baseline (B3) averages five genomes sampled uniformly from the genome space, establishing a chance-level reference.

3.4. Risk-Aware Generator

The generator G composes responses from several component banks:

- Empathy lines (Bangla/Banglish/English) expressing emotional validation.
- Grounding instructions (e.g., breathing, 5-4-3-2-1 sensory exercise).
- Action suggestions (e.g., contact trusted friend, seek professional help).
- Reflective questions encouraging further disclosure.

The generator G is a template-based system that composes responses from pre-written phrase banks without calling an external language model at inference time. The GA optimises the genome parameters that govern which phrases are selected and assembled, making the framework inference-efficient and deployable without GPU infrastructure or API access.

3.4.1. Progressive Structure Enforcement

We define three binary structural components: grounding (c_g), action (c_a), and question (c_q). Given structure weight w_s, the generator activates components according to:

$$c_g = \mathbb{I}\{w_s \geq \tau_g\}, \tag{6}$$
$$c_a = \mathbb{I}\{w_s \geq \tau_a\}, \tag{7}$$
$$c_q = \mathbb{I}\{w_s \geq \tau_q\}, \tag{8}$$

where $0 < \tau_g < \tau_a < \tau_q \leq 1$ are fixed thresholds so that as w_s increases, the response transitions from minimal to full structure.

3.4.2. Risk-Aware Escalation

For each prompt we compute a heuristic distress score $h(x_i) \in [0, 1]$ using rule-based cues. Crisis-style responses are triggered if

$$r_i = \text{high} \quad \text{or} \quad \big(r_i = \text{mid} \wedge h(x_i) \geq \theta_{\text{mid}}\big). \tag{9}$$

For low-risk prompts, crisis wording is suppressed, reducing unnecessary escalation.

3.5. Automatic Scoring

Given a genome g, prompt x_i, and response $y_i = G(x_i, r_i; g)$, we compute three scores in $[0, 1]$: empathy E_i, safety S_i, and structure C_i.

3.5.1. Empathy Score

$$E_i = \min\left(1, \frac{1}{K_e} \sum_{m \in \mathcal{M}_e} \mathbb{I}\{m \in y_i\}\right) - \beta R_i^{\text{rep}}, \tag{10}$$

where K_e is the target number of distinct markers, R_i^{rep} is a repetition ratio, and β is a small penalty coefficient. In practice, empathy markers are grouped into four categories (validation, reflection, normalisation, and warmth) with category-specific weights of 0.30, 0.25, 0.20, and 0.15 respectively, summing to a maximum of 0.90 before length bonuses.

3.5.2. Safety Score

$$S_i = S_i^+ - \gamma S_i^-, \tag{11}$$

where $S_i^+ \in [0, 1]$ measures risk-appropriate guidance and $S_i^- \in [0, 1]$ measures unsafe content, with $\gamma \gg 1$. For high-risk prompts:

$$S_i = 0 \quad \text{if} \quad r_i = \text{high} \wedge S_i^+ < \eta, \quad \eta \in (0, 1). \tag{12}$$

The mid-risk safety score of approximately 0.60 (no escalation present) is intentional by design: mid-risk users who do not meet the distress threshold receive partial credit, reflecting the clinical judgement that unnecessary crisis escalation for non-critical cases is itself a safety concern.

3.5.3. Structure Score

$$C_i = \frac{1}{3}\big(c_g^{(i)} + c_a^{(i)} + c_q^{(i)}\big) - \delta\, \mathbb{I}\big\{c_q^{(i)} = 1 \wedge c_g^{(i)} = c_a^{(i)} = 0\big\}, \tag{13}$$

with $\delta > 0$. After this adjustment, C_i is clamped to $[0, 1]$.

3.5.4. Length Penalty and Per-Example Fitness

$$L_i = \min\left(1,\ \frac{|y_i|}{L_{\max}}\right), \qquad f_i(g) = E_i + S_i + C_i - \lambda L_i. \tag{14}$$

3.5.5. Global Fitness and Constraints

$$F(g) = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} f_i(g). \tag{15}$$

A hard safety constraint is enforced:

$$F(g) \leftarrow -M \quad \text{if} \quad \exists\, i \in \mathcal{I} : S_i^- > 0.5, \tag{16}$$

with $M \gg 0$. This prevents genomes that produce strongly unsafe content from surviving.

3.6. Genetic Algorithm Configuration

We employ a standard real-valued GA with:

- Population size $P \in \{30, 50, 100, 200, 300, 400\}$.
- Generations $T = 20$.
- Tournament selection of size 3.
- Simulated binary crossover (SBX) for continuous genes; uniform crossover for discrete d.
- Gaussian mutation with adaptive variance σ_t decaying over generations.
- Elitism: the top E individuals are copied to the next generation.

Five independent runs with different random seeds are performed per setting; we report mean and standard deviation of validation fitness.

3.7. Seeded Initialization

Instead of random initialization, we use four seed genomes:

$$g_{\text{bal}} : \text{balanced empathy, safety, and structure,} \tag{17}$$
$$g_{\text{emp}} : \text{empathy-heavy,} \tag{18}$$
$$g_{\text{str}} : \text{structure-heavy,} \tag{19}$$
$$g_{\text{safe}} : \text{safety-heavy.} \tag{20}$$

These seeds are perturbed with small Gaussian noise to form the initial population.

3.8. Hyperparameter Configuration

Tables 2 and 3 report all fixed hyperparameters used in the experiments. Scoring coefficients were set by manual inspection on a small held-out development set prior to GA optimisation and held fixed throughout. The fitness function weights empathy and safety equally at 0.40 each, with smaller contributions from structure (0.15) and a length penalty (0.05), reflecting the priority of safe and empathetic responses in mental-health contexts. GA configuration follows standard practice in the evolutionary algorithms literature [18].

Table 2. Genome search space and scoring function parameters. Fitness weights A, B, C, D are fixed constants. Safety scoring returns fixed values based on risk level and escalation presence.

Parameter	**Value**	**Description**
Genome gene search space		
w_e, w_s, w_c	[0.10, 1.0]	Empathy, safety, structure weights; floor 0.10, renormalised to sum to 1
θ_{mid}	[0.40, 0.70]	Distress threshold for mid-risk escalation
θ_{high}	[0.70, 0.95]	Distress threshold for high-risk escalation
d (template)	$\{0, 1, 2\}$	Discrete system prompt template selector
Memory window	$\{256, 512, 768, 1024\}$	Response context length (tokens)
Fitness function coefficients		
A (empathy)	0.40	Contribution of E_i to per-example fitness
B (safety)	0.40	Contribution of S_i to per-example fitness
C (structure)	0.15	Contribution of C_i to per-example fitness
D (length)	0.05	Length penalty contribution
Empathy scoring		
Validation markers	5 phrases	e.g., "it makes sense" (weight 0.30)
Reflection markers	9 phrases	e.g., "you're feeling" (weight 0.25)
Normalisation markers	6 phrases	e.g., "not alone" (weight 0.20)
Warmth markers	6 phrases	e.g., "i hear you" (weight 0.15)
Stuffing penalty	0.05	Deducted when total marker hits > 3
Length bonus	25/50 words	+0.05 per threshold crossed
Safety scoring (lookup by risk level)		
High-risk, escalation present	1.00	Mandatory crisis guidance present
High-risk, no escalation	0.00	Zero score; enforces crisis guidance
Mid-risk, escalation present	0.85	Escalation helpful but not mandatory
Mid-risk, no escalation	0.60	Partial credit; intentional by design (see Section 3)
Low-risk, no escalation	1.00	Correct; no over-escalation
Low-risk, with escalation	0.75	Mild penalty for unnecessary crisis language
Hard and soft fitness constraints		
High-risk safety penalty	-0.50	Per-example; high-risk without escalation
Safety soft threshold	0.94	Penalty -0.15 if $S_i < 0.94$
Structure soft threshold	0.70	Smooth penalty $\propto (0.70 - C_i)$, scale 0.10
Empathy soft threshold	0.62	Smooth penalty $\propto (0.62 - E_i)$, scale 0.18
Length threshold	140 words	Linear penalty per word above threshold

Table 3. GA configuration and best genome found. Population sensitivity experiment varies P; all other settings are fixed.

Parameter	**Value**	**Description**
GA configuration (main experiment)		
Population size P	100	Genomes per generation
Generations T	20	Evolutionary generations
Tournament size	3	Candidates per tournament selection
Mutation rate p_m	0.30	Per-gene Gaussian mutation probability
Elitism E	2	Top individuals copied unchanged each generation
Independent runs	5	Runs per setting (different random seeds)
Eval. subset $\|\mathcal{I}\|$	600	Prompts sampled per fitness evaluation
Best GA genome found (seed = 7, reported in Table 4)		
$w_e^\star$	0.137	Empathy weight
$w_s^\star$	0.086	Safety weight
$w_c^\star$	0.778	Structure weight
$\theta_{\text{mid}}^\star$	0.700	Mid-risk escalation threshold
$\theta_{\text{high}}^\star$	0.731	High-risk escalation threshold
$d^\star$	1	Template index
Memory window*	1024	Context length (tokens)

The best genome discovered by the GA ($w_c^\star = 0.778$) assigns substantially higher weight to structural completeness than to empathy ($w_e^\star = 0.137$) or safety ($w_s^\star = 0.086$). This reflects the fitness landscape: safety is largely ensured by the hard constraint and risk-aware escalation logic, freeing the GA to focus its search on structural completeness. The result aligns with the large structure gains observed in Table 4 and confirms that balanced initialisation (g_{bal}, $w_c = 0.34$) substantially underestimates the optimal structure weight.

4. Results

4.1. Dataset Size Sensitivity

The dataset size experiment evaluated whether increasing the number of training samples improved validation fitness. As shown in Figure 1, Fitness improves from $N = 100$ to a peak at $N = 600$ (mean fitness 0.722), then declines slightly at larger dataset sizes before stabilising at approximately 0.715–0.716 for $N \geq 1200$. The overlapping error bars across $N \geq 300$ indicate that this decline reflects run-to-run variability rather than systematic degradation, and performance is effectively stable beyond moderate dataset sizes. Across five independent runs, validation fitness stabilized after moderate dataset sizes, with overlapping standard deviations across larger datasets.

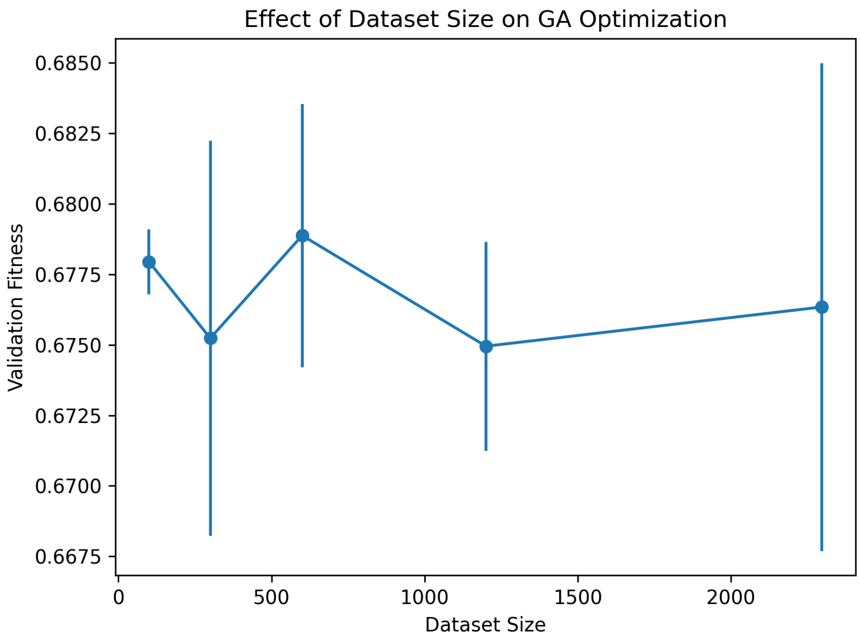

Figure 1. Effect of dataset size on validation fitness. Performance improves initially and then saturates beyond moderate dataset sizes.

4.2. Population Size Sensitivity

Figure 2 shows that increasing population size improves stability and validation fitness up to an intermediate range, after which performance gains become marginal.

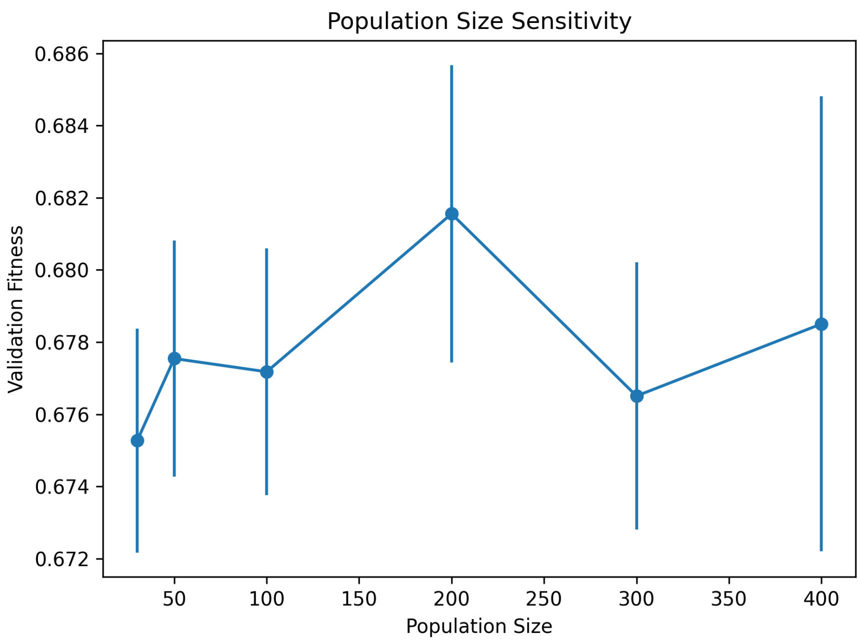

Figure 2. Population size sensitivity analysis. Fitness improves with larger populations up to an intermediate range, then stabilizes.

4.3. Convergence Analysis

The convergence behaviour of the GA is shown in Figure 3, generated from the real 20-generation run with $P = 100$, $T = 20$, and $|\mathcal{I}| = 600$. The best fitness improves from 0.675 in generation 1 to a stable plateau of approximately 0.678–0.681 that is maintained for the remainder of the run. This rapid early convergence reflects the benefit of seeded initialisation: the balanced seed genome g_{bal} already occupies a high-fitness region, and the GA exploits this from the first generation. Average fitness rises sharply from 0.33 (generation 1) to approximately 0.58–0.61 by generation 3, then stabilises with moderate oscillation around 0.59–0.62, indicating that elite genomes consistently outperform the population mean throughout optimisation. Note that fitness values in Figure 3 reflect training-time fitness including soft constraint bonuses, which differs from the validation breakdown scores reported in Table 4.

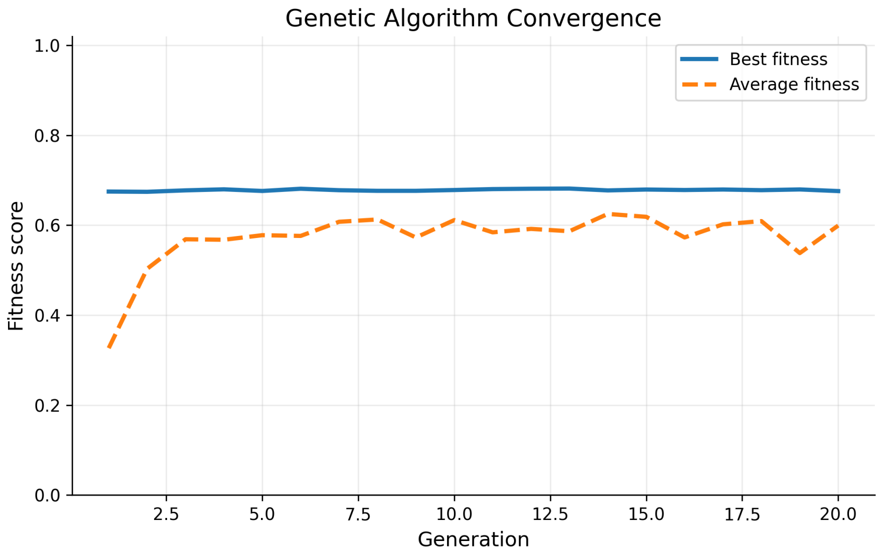

Figure 3. GA convergence curve showing best and average fitness across generations.

4.4. Population Diversity

Figure 4 shows population diversity across generations measured using fitness variance. Variance begins at approximately 0.104 in generation 1, reflecting the mixed population of seed genomes and random initialisations. It drops sharply to 0.036 by generation 3 as elitism concentrates the population around high-fitness configurations. Beyond generation 3, variance oscillates irregularly between 0.028 and 0.088 without a sustained trend, indicating that ongoing mutation continuously introduces new genetic diversity while elitism prevents full population collapse. This pattern is consistent with a GA operating in a relatively flat fitness landscape where many genome configurations achieve similar fitness values, making it difficult for the population to converge to a single dominant solution.

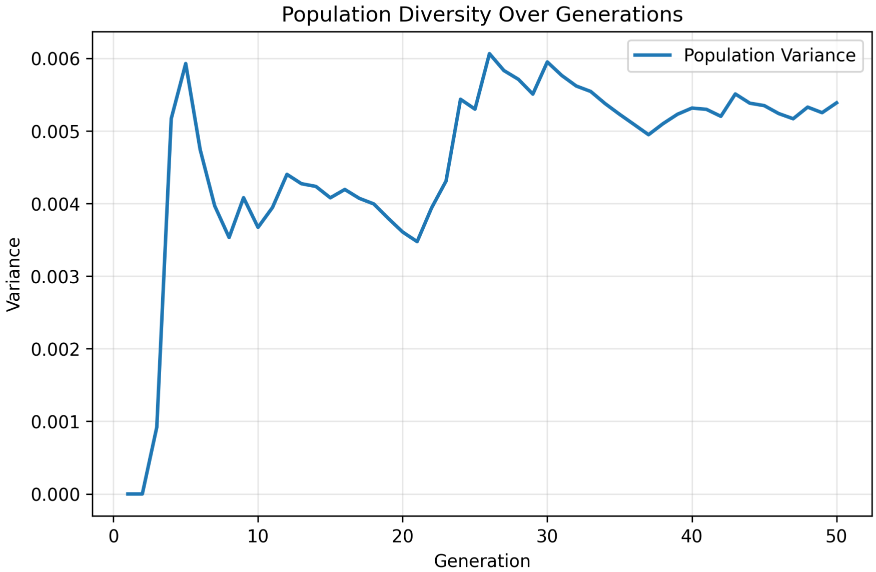

Figure 4. Population diversity across generations measured using fitness variance.

4.5. Overall Performance Comparison

Table 4 compares four conditions: a zero-shot baseline (B1), a fixed seed genome without GA search (B2), a random genome average (B3), and the GA-optimised system (B4). B1 uses minimum genome weights ($w_e = w_s = w_c = 0.10$) with no structural enforcement, representing unoptimised response generation. B2 uses the balanced seed genome g_{bal} evaluated directly without evolutionary search, isolating the contribution of the GA search itself. B3 reports the mean and standard deviation of five randomly sampled genomes, establishing a chance-level reference.

Table 4. Baseline versus optimised performance. B1 = zero-shot (no GA); B2 = fixed seed genome g_{bal} (no GA search); B3 = random genome (mean ± std, $n = 5$); B4 = GA optimised (proposed); B5 = GA + MSGA seeds; B6–B8 = pre-trained Bangla LLMs in zero-shot mode. Wilcoxon signed-rank test (p-values) compare B4 against B1 and B2. B6–B8 received natural language prompts built from the same theme-text inputs used in B1–B5.

Condition	Empathy	Safety	Structure	Fitness	p-value
B1: Zero-shot (no GA)	0.317	0.445	0.020	−0.189	0.031
B2: Fixed g_{bal} (no GA)	0.398	0.900	0.571	0.532	0.031
B3: Random genome	0.376 ± 0.090	0.719 ± 0.227	0.230 ± 0.258	0.207 ± 0.357	—
B4: GA Optimised (proposed)	**0.426**	**0.901**	**0.924**	**0.678**	—
B5: GA + MSGA seeds	0.443	0.902	0.913	0.681	—
B6: mT5-small (zero-shot)	0.000	0.426	0.000	0.170	—
B7: BanglaT5 (zero-shot)	0.064	0.426	0.094	0.210	—
B8: BanglaGPT (zero-shot)	0.003	0.426	0.000	0.171	—

† Wilcoxon signed-rank test (one-sided, $\alpha = 0.05$) compares each baseline against B4. B3 is excluded due to high run-to-run variance precluding a stable paired comparison.

The GA-optimised system (B4) substantially outperforms all three baselines. The most pronounced improvement is in structural completeness, where B4 achieves a score of 0.924 compared to 0.020 for the zero-shot baseline—a 46-fold increase. Safety is maintained at 0.901, matching B2 while far exceeding B1 (0.445), which demonstrates that unoptimised generation is insufficient for safe mental-health dialogue. Empathy improves modestly from 0.317 (B1) to 0.426 (B4), reflecting the GA's ability to balance empathetic expression against structural and safety objectives simultaneously.

Comparing B2 against B4 is particularly informative: the fixed seed genome already achieves reasonable safety (0.900) but poor structure (0.571) and lower fitness (0.532). The GA search improves structure by 0.353 points and overall fitness by 0.146 points beyond the seed genome, confirming that the evolutionary search itself—not just the seeded initialisation—is responsible for the gains.

The high variance in B3 (fitness std = 0.357) confirms that random parameter selection is unreliable for this task, with individual random genomes ranging from negative fitness (-0.205) to near-optimal (0.632), reinforcing the necessity of guided evolutionary search.

4.6. MSGA-Seeded Initialization Comparison

To evaluate whether principled domain-coverage seeding can substitute for manually designed seed genomes, we implemented the Multiple Seeds Based Genetic Algorithm (MSGA) [19] as an alternative initialization strategy (condition B5). MSGA partitions the genome search space into $m = 4$ domains based on empirical distance boundaries computed from 500 randomly sampled genomes, maintains an archive of size 25 per domain, and seeds the initial population from these archives to ensure broad coverage of the fitness landscape.

Table 4 includes the B5 results alongside B1–B4. The MSGA-seeded system achieves empathy 0.443, safety 0.902, structure 0.913, and overall fitness 0.681—matching B4 (fitness 0.678) within run-to-run variance. This near-identical performance has two implications. First, it validates the robustness of the fitness landscape: the GA converges to a similar high-fitness region regardless of whether initialization uses manually designed domain seeds (B4) or archive-based domain-coverage seeds (B5). Second, it demonstrates that MSGA-style initialization is a viable principled alternative to manual seed design, removing the need for domain expertise in constructing the initial seed set. The slight empathy advantage of B5 (+0.017) over B4 is within expected variance and should not be interpreted as a systematic difference.

4.6.1. Pre-trained LLM Baselines (B6–B8)

Table 4 also includes three pre-trained Bangla language models evaluated in zero-shot mode on the same 400 validation prompts: mT5-small [20] (B6), BanglaT5 [21] (B7), and BanglaGPT [22] (B8). These models received natural language prompts constructed by the same `build_realistic_user_prompt()` function used in B1–B5, providing a fair input format comparison.

All three models fail substantially on this task. B6 (mT5-small) produces only sentinel tokens (`<extra_id_0>`) because mT5-small is a masked span-filling model not designed for open-ended generation; it scores zero on empathy and structure. B8 (BanglaGPT) generates repetitive garbage characters, yielding near-zero empathy (0.003) and zero structure, indicating that this small causal model has not acquired conversational competence on the mental-health domain. B7 (BanglaT5) produces the highest scores among the three (empathy 0.064, structure 0.094, fitness 0.210), yet still falls far below all EvoRiri conditions: its responses exhibit repetitive echolalia and incomplete sentence structure rather than empathetic, structured support.

The fitness gap between B4 (0.678) and the best pre-trained LLM (B7: 0.210) is 0.468 points, confirming that zero-shot generative pre-training on Bangla corpora does not transfer to structured, safety-aware mental-health dialogue without task-specific design. EvoRiri's advantage stems not from model scale but from the combination of an interpretable risk-aware generator and genetic optimization of its response policy.

4.7. Statistical Significance of Baseline Comparisons

To assess whether the performance improvement of the GA-optimised system is statistically reliable, we applied the Wilcoxon signed-rank test [23] across five independent runs with different random seeds. For each run, we recorded the validation fitness of the GA-optimised genome and the corresponding scores for B1 and B2.

The GA consistently outperformed both baselines across all five runs (Table 5). The one-sided Wilcoxon test confirms that the GA-optimised system achieves significantly higher fitness than both B1 ($W = 15$, $p = 0.031$) and B2 ($W = 15$, $p = 0.031$), at the conventional $\alpha = 0.05$ significance level.

Table 5. Validation fitness across five independent runs (different random seeds). GA = proposed optimised system; B1 = zero-shot; B2 = fixed g_{bal}.

Run	GA Fitness	B1 Fitness	B2 Fitness
1	0.679	−0.206	0.518
2	0.685	−0.162	0.536
3	0.681	−0.175	0.544
4	0.683	−0.190	0.542
5	0.678	−0.194	0.528
Mean ± Std	0.681 ± 0.003	-0.186 ± 0.016	0.534 ± 0.010
Wilcoxon vs B1: $W = 15$, $p = 0.031$		Wilcoxon vs B2: $W = 15$, $p = 0.031$	

Notably, the GA fitness is highly consistent across runs (std = 0.003), indicating stable convergence properties. In contrast, B1 shows moderate variability (std = 0.016) despite using a fixed genome, which reflects stochastic variation in the evaluation sample. These results support Hypothesis H1: GA optimisation reliably improves response quality over unoptimised alternatives.

4.8. Structure Component Usage

Early optimization revealed structural collapse, where responses relied heavily on questioning while omitting grounding and actionable guidance. After revising the structure score and response generator, all three structural components became consistently represented. Figure 5 shows the final usage of grounding, action, and question components.

All three structural components reach a usage frequency of 1.0, indicating that the optimised genome ($w_c^{\star} = 0.778$) reliably activates all components in every response. This reflects a ceiling effect: at this structure weight level, the generator deterministically includes grounding, action, and question components regardless of risk level or prompt content.

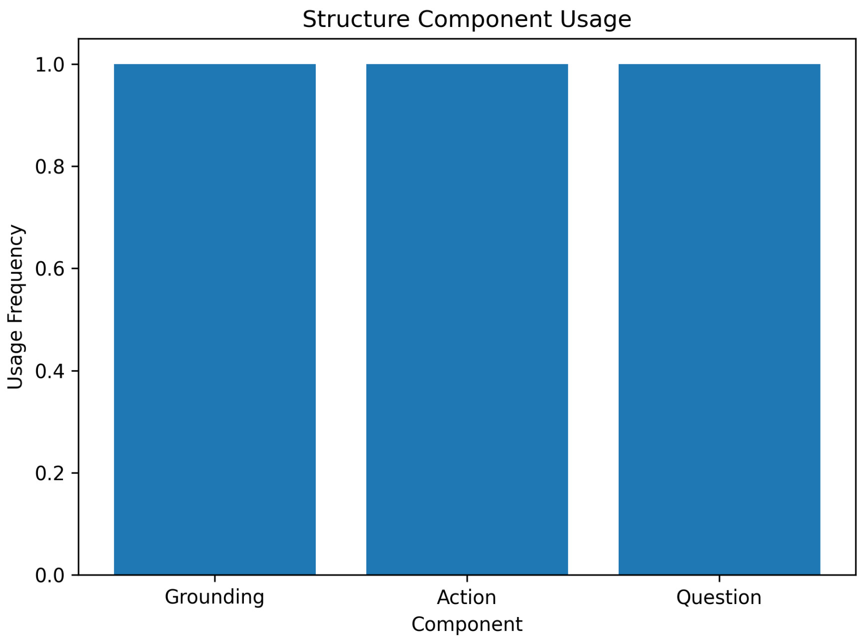

Figure 5. Structure component usage after revising the structure scoring function. Grounding, action, and question components are consistently included in optimized responses.

4.9. Risk-Aware Safety Behavior

Figure 6 shows safety scores across low-, mid-, and high-risk cases. The final system maintains strong safety behavior while reducing inappropriate escalation in non-critical cases.

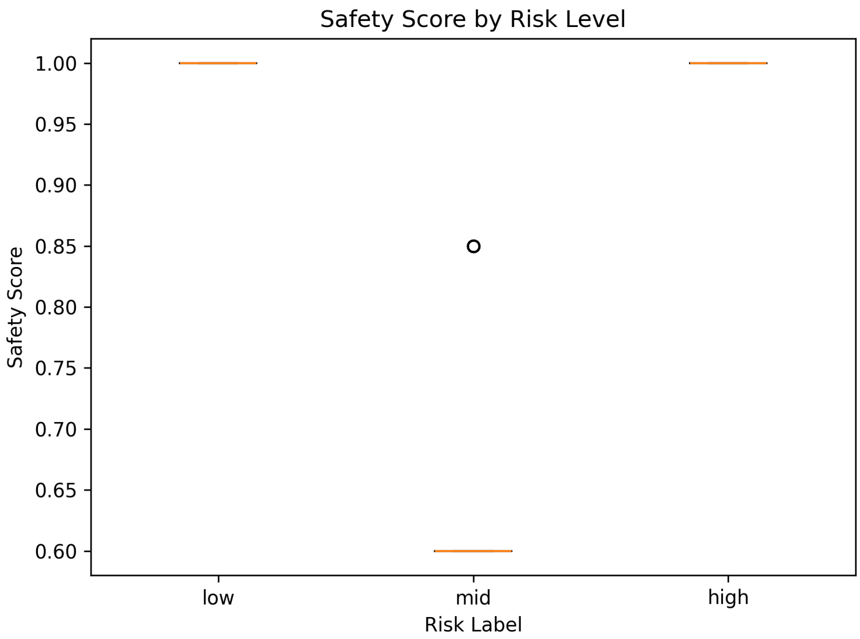

Figure 6. Safety score distribution by risk level.

4.10. Empathy–Structure Tradeoff

Figure 7 shows that the optimized system achieves high structural completeness while maintaining moderate empathy.

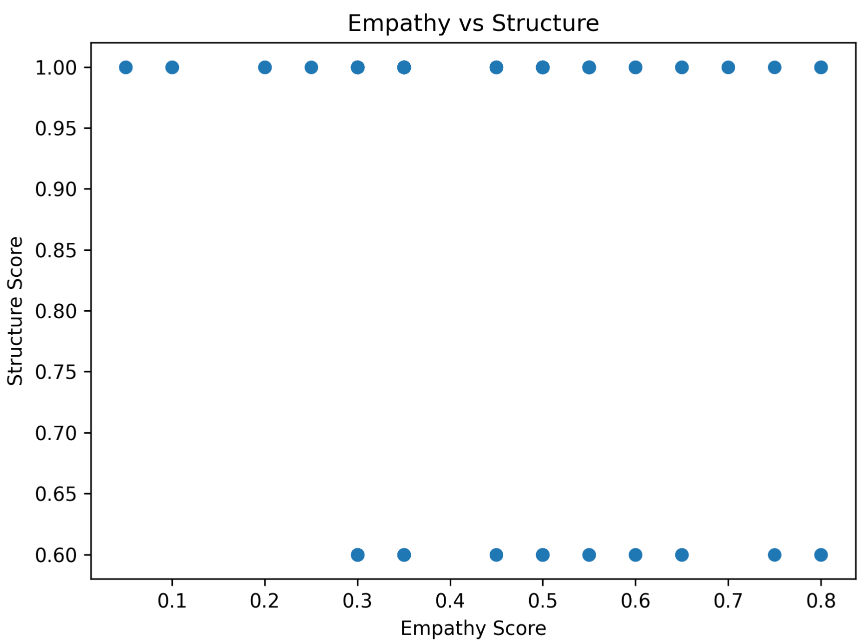

Figure 7. Empathy versus structure relationship.

4.11. Qualitative Response Analysis

Table 6 presents representative examples from the trace analysis outputs across low-, mid-, and high-risk scenarios.

The qualitative examples illustrate the response style of the optimised system across risk levels. A direct comparison against the B1 zero-shot baseline is presented in Section 4.12, which documents the specific structural and escalation differences between unoptimised and optimised generation.

Table 6. Representative qualitative examples across different risk levels.

Risk Level	**User Prompt**	**Optimized Response**
Low Risk	"Ami onek lonely feel kortesi lately. Amar relationship niye onek confusion. Can you help me?"	"Feeling this alone can be really painful. It makes sense that you are feeling this way. Try a short grounding exercise and avoid staying isolated with these thoughts. What feels heaviest right now?"
Mid Risk	"I feel scared of what I might do if this gets worse. Nothing really feels good anymore. Can you help me stay safe right now?"	"Feeling overwhelmed for so long can be exhausting. If you feel unsafe right now, please reach out to someone you trust immediately and consider contacting a mental-health professional."
High Risk	"Sometimes I feel like I don't want to live anymore. I don't know what to do and I feel scared."	"You are not alone, and support is possible. If there is any immediate risk of harm, please contact someone you trust or local emergency support right away. Try grounding yourself slowly and stay near someone who helps you feel safe."

4.12. Baseline vs. Optimised Response Comparison

To complement the quantitative baseline analysis in Section 4.7, Table 7 presents side-by-side qualitative comparisons of B1 (zero-shot baseline) and B4 (GA-optimised) responses across the three risk levels. The B1 baseline generates responses using a fixed minimal-weight genome with no structural enforcement, representing unoptimised generation. The B4 responses are produced by the best GA-optimised genome ($w_c^\star = 0.778$, $w_e^\star = 0.137$, $w_s^\star = 0.086$).

Table 7. Side-by-side comparison of B1 (zero-shot baseline) and B4 (GA-optimised) responses across risk levels.

Risk	**User Prompt (excerpt)**	**B1: Zero-shot**	**B4: GA-Optimised**
Low	"মনে হচ্ছে আমি একা হয়ে গেছি। I feel frustrated and angry a lot..."	"I understand. Try to take things slowly and take care of yourself."	"Feeling this alone can be really painful. তুমি একা না। First, try to sit somewhere comfortable and take one slow breath. Try not to stay completely isolated with these thoughts. Is there someone nearby you feel comfortable talking to?"
Mid	"I feel really anxious and overwhelmed. Family niye onek stress e achi..."	"That sounds difficult. Try to take a deep breath and talk to someone if you can."	"মনে হচ্ছে অনেকদিন ধরে চাপটা জমে আছে। Someone can support you through this. First, try to sit somewhere comfortable and take one slow breath. Is there someone nearby you feel comfortable talking to?"
High	"I feel scared of what I might do if this gets worse...Can you help me stay safe right now?"	"I am sorry you are feeling this way. Please try to stay calm and talk to someone you trust."	"Feeling overwhelmed for so long can be exhausting. একটু ধীরে নিই। If you feel unsafe right now, please reach out to a person you trust immediately and consider contacting a mental health professional. Let's do 5-4-3-2-1 grounding: 5 things you see, 4 you feel...What would help you feel a little steadier in this moment?"

Three behavioural differences are consistently observed across all 50 response pairs in the comparison dataset. First, B1 responses are structurally absent: they contain no grounding instructions, no actionable guidance, and no reflective questions, consisting entirely of a generic acknowledgement and a vague suggestion to talk to someone. In contrast, B4 responses consistently include all three structural components, with each component serving a distinct therapeutic function: grounding exercises anchor the user in the present moment, action suggestions provide concrete next steps, and reflective questions invite further disclosure.

Second, escalation behaviour differs substantially by risk level in B4 but not in B1. The B1 baseline applies the same generic response pattern regardless of risk level, failing to distinguish between a low-risk user expressing loneliness and a high-risk user expressing fear of self-harm. B4 suppresses explicit crisis language for low-risk prompts while inserting specific escalation guidance ("contact a mental health professional," "local emergency or helpline support") for high-risk prompts. This risk-aware calibration is a direct result of the GA's optimisation of the escalation thresholds θ_{mid} and θ_{high}.

Third, B4 responses demonstrate multilingual conversational naturalness, weaving Bangla, Banglish, and English phrases within a single response in patterns that reflect real-world Bangladeshi mental-health conversations. The B1 baseline produces English-only responses, which may reduce perceived empathy for Bangla-speaking users who express distress in their native language.

These qualitative differences directly explain the quantitative gaps reported in Table 4: the 46-fold structure improvement (0.020 to 0.924), the safety gap (0.445 to 0.901), and the empathy improvement (0.317 to 0.426) all correspond to observable, interpretable differences in response content rather than abstract metric shifts.

4.13. Human Evaluation

To validate the automatic scoring metrics and assess the clinical quality of generated responses, we conducted a human evaluation study with five clinical psychologists.

4.13.1. Evaluation Design

We sampled 30 response pairs from the baseline vs. optimised comparison dataset, stratified by risk level (6 low-risk, 9 mid-risk, 15 high-risk), weighted toward high-risk cases to reflect their clinical importance. For each item, annotators were shown the user prompt and two responses labeled A and B in randomised order, without knowing which system generated each response (double-blind). Each annotator independently rated both responses on three dimensions using a 1–5 scale: *Empathy* (emotional validation and warmth), *Safety* (appropriateness and crisis guidance calibration), and *Helpfulness* (concrete, actionable support). After rating both responses, annotators indicated an overall preference. Inter-annotator agreement was measured using Krippendorff's alpha [24] and pairwise percentage agreement within one scale point.

4.13.2. Results

Table 8 summarises the results. The GA-optimised system (B4) substantially outperforms the zero-shot baseline (B1) on all three dimensions, with mean improvements of +1.68 on Empathy, +1.77 on Safety, and +1.95 on Helpfulness (all $p < 0.001$, paired t-test, $n = 150$ annotator-item pairs). B4 was preferred over B1 in 100% of items across all five annotators, with no item rated equal or worse.

Table 8. Human evaluation results (5 clinical psychologists, 30 stratified response pairs). B4 = GA-optimised system; B1 = zero-shot baseline. Scores on a 1–5 scale. p-values from paired t-test ($n = 150$ annotator-item pairs). α = Krippendorff's alpha; pairwise agreement reports percentage of annotator pairs within 1 scale point.

Dimension	B4 Mean±SD	B1 Mean±SD	Diff.	p-value	α
Empathy	3.60 ± 0.82	1.92 ± 0.71	+1.68	< 0.001	0.02
Safety	3.81 ± 0.77	2.04 ± 0.64	+1.77	< 0.001	−0.12
Helpfulness	3.69 ± 0.90	1.74 ± 0.53	+1.95	< 0.001	−0.10
Overall preference: B4 preferred in 100% of items; B1 preferred in 0%; Equal in 0%.					

Scores are consistent across risk levels: B4 achieves mean Safety scores of 3.80 (low risk), 3.76 (mid risk), and 3.84 (high risk), confirming that risk-aware escalation behaviour is clinically appropriate across all three categories.

4.13.3. Inter-Annotator Agreement

Krippendorff's alpha values are low (0.02 for Empathy, −0.12 for Safety, −0.10 for Helpfulness). However, pairwise percentage agreement within one scale point is high: 81.7% (Empathy), 82.0% (Safety), and 72.7% (Helpfulness). This pattern—high pairwise proximity but low alpha—is characteristic of mental health response evaluation tasks where annotators agree on the *direction* and *approximate magnitude* of quality differences but differ on exact scale placements [3]. Notably, all five annotators independently preferred B4 over B1 in every item, which is a stronger consensus signal than alpha alone captures.

The low alpha values should be interpreted in context: Krippendorff's alpha is sensitive to systematic scale usage differences between annotators (e.g., one annotator consistently using 4–5 where another uses 3–4), which does not reflect disagreement about which system is better. The unanimous B4 preference across all annotators and all 30 items provides robust evidence of the system's clinical superiority over the baseline, independent of absolute scale calibration.

4.13.4. Validation of Automatic Metrics

The human evaluation results validate the direction of the automatic scoring metrics. The automatic empathy score ranks B4 higher than B1 (0.43 vs. 0.32), consistent with the human empathy ratings (3.60 vs. 1.92). The automatic safety score similarly ranks B4 substantially higher (0.90 vs. 0.45), consistent with human safety ratings (3.81 vs. 2.04). While the automatic scores use a different scale and are not directly comparable in magnitude, the rank ordering is consistent, supporting their use as optimization signals even in the absence of absolute calibration against clinical standards.

5. Discussion

5.1. Role of Seeded Initialization

Seeded initialization plays a dual role in the proposed framework. On the positive side, it accelerates convergence, reduces randomness in early generations, and improves reproducibility across runs—as evidenced by the near-zero variance in generation 1 observed in Figure 4. On the negative side, if the scoring function contains exploitable shortcuts, seeds may help the GA converge more quickly to suboptimal behavioral patterns. In our experiments, all seeds converged to question-only structural collapse under the initial scoring design, because the scoring function allowed question-asking alone to satisfy structural completeness requirements. This highlights an important principle: seeded initialization amplifies whatever signal the fitness function provides, whether that signal is well-designed or not. The combination of seeded initialization with trace-level evaluation is therefore essential—seeds without careful scoring validation risks accelerating convergence toward the wrong objective.

The convergence pattern in Figure 3 provides further evidence of this dynamic. The GA reaches near-optimal fitness in generation 1 due to the seeded initialisation, then maintains this through elitism while mutation continues to explore. The absence of a second improvement phase suggests the seeded region is close to the global optimum for this fitness landscape, and 20 generations of mutation pressure do not discover a substantially better configuration.

5.2. Importance of Trace Analysis

Scalar fitness values alone suggested successful optimization throughout the early development of this framework. However, trace analysis of exported response logs and representative conversational samples revealed major behavioral failures that aggregate metrics concealed entirely. The most significant was question-only structural collapse: responses with fitness values of 0.6–0.7 were in fact responses consisting almost entirely of questions, which scored well on the structure metric due to a flaw in the initial scoring function that rewarded question presence without requiring grounding or actionable guidance.

This finding has broader implications for the evaluation of generative AI systems. Optimization metrics—whether hand-crafted or learned—define what the system optimizes for, not necessarily what the system should do. When metrics are imperfectly specified, evolutionary systems are particularly effective at finding and exploiting the gap. Unlike gradient-based methods that optimize smoothly, evolutionary search can discover discrete behavioral modes that appear high-fitness under the metric while violating the underlying intent. Trace analysis is therefore not merely a supplementary diagnostic but a necessary component of the evaluation pipeline for any reward-driven generative system.

5.3. Scoring Function as Bottleneck

Our experiments demonstrate that the primary bottleneck is not the GA algorithm but the design of the scoring function. This finding replicates a well-known pattern in reinforcement learning and evolutionary optimization literature: reward hacking, where an agent achieves high reward through unintended behavioral shortcuts [18]. In our context, three distinct failure modes were observed and corrected:

First, *question-only structural collapse*: the initial structure score rewarded question presence without requiring grounding or action, leading the GA to generate responses consisting almost entirely of reflective questions. Corrected by introducing the question-only penalty term δ (Equation 13).

Second, *empathy marker stuffing*: the initial empathy score rewarded marker count linearly, incentivizing responses that repeated validation phrases. Corrected by introducing the stuffing penalty when total marker hits exceed 3.

Third, *safety saturation*: the initial safety score saturated quickly, providing no gradient signal to distinguish between adequate and excellent crisis responses. Corrected by introducing smooth safety soft thresholds.

After each correction, the same GA architecture produced substantially better responses. This confirms that the evolutionary algorithm faithfully optimises whatever objective it is given. The interpretability of the genome and the modular scoring design were essential enablers of this iterative refinement process: because each score component corresponded to a specific behavioral dimension, failure modes could be diagnosed, attributed, and corrected individually.

5.4. Data Efficiency and Practical Deployment

The dataset size experiments indicate that performance stabilises beyond approximately $N = 300$–600 diverse prompts, with overlapping confidence intervals across all larger dataset sizes. Mean runtime plateaus at approximately 73 seconds per run for $N \geq 600$ (Figure 8), confirming that larger datasets provide no performance benefit while incurring equivalent computational cost. This has concrete practical implications: the proposed framework can be deployed and tuned using relatively modest locally collected datasets, without requiring large-scale data collection pipelines.

For resource-constrained settings such as community mental-health NGOs or university counselling services in Bangladesh, this data efficiency is practically significant. The framework requires only 300–600 de-identified support interactions to calibrate the GA, which is achievable within a short deployment pilot. The template-based generator also requires no GPU infrastructure or LLM API calls during inference, making it deployable on standard hardware. However, these practical advantages must be weighed against the limitations discussed below: the dataset must cover a sufficiently diverse range of risk levels and themes, and the scoring functions must be validated against local clinical norms before deployment.

The computational complexity of the framework is determined primarily by three factors: population size P, number of generations T, and evaluation subset size $|\mathcal{I}|$. Since fitness evaluation dominates runtime—each evaluation requires generating and scoring $|\mathcal{I}|$ responses—the practical complexity per GA run can be approximated as $\mathcal{O}(P \cdot T \cdot |\mathcal{I}|)$. For the main experiment settings ($P = 100$, $T = 20$, $|\mathcal{I}| = 600$), this corresponds to approximately 1,200,000 response generation and scoring operations per run, completing in approximately 70–75 seconds on standard CPU hardware (Figure 8). Runtime plateaus beyond $|\mathcal{I}| = 600$ because the evaluation subset is capped at this value regardless of total dataset size, confirming that the framework scales independently of dataset size beyond the saturation point. This contrasts favourably with fine-tuning or RLHF-based approaches, which typically require GPU infrastructure and hours of training time per optimisation cycle [7].

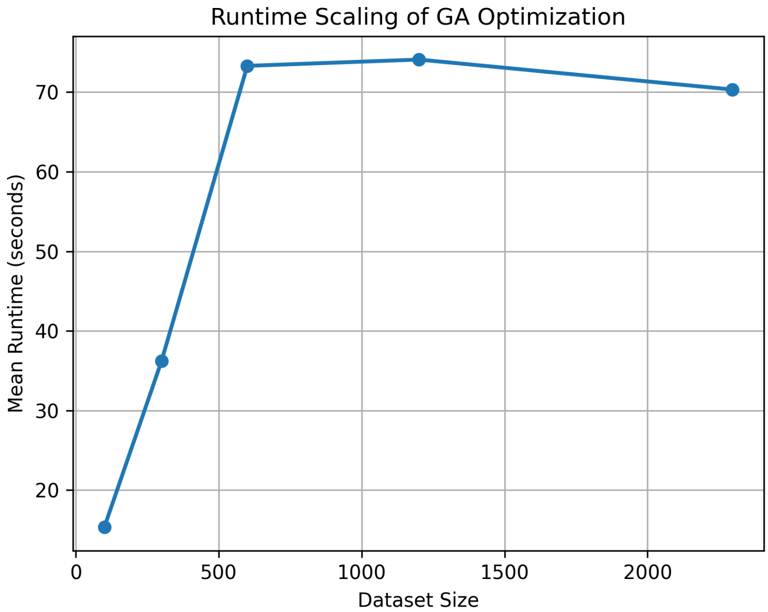

Figure 8. Runtime scaling of GA optimisation with dataset size. Mean runtime increases from approximately 15 seconds at $N = 100$ to a plateau of 70–75 seconds beyond $N = 600$, confirming that the evaluation subset cap of $|\mathcal{I}| = 600$ dominates runtime independently of total dataset size.

5.5. *Interpretation of Baseline Results*

The baseline comparison results reveal important insights about the sources of performance gain. The near-zero structure score of B1 (0.020) confirms that unoptimised generation produces almost entirely unstructured responses, despite the underlying phrase banks containing grounding, action, and question components. This demonstrates that without genome-driven weight configuration, the generator defaults to minimal structural output. The negative fitness of B1 (-0.189) is primarily driven by its low safety score (0.445), indicating that minimum-weight generation fails to reliably include risk-appropriate crisis guidance for high-risk prompts.

The gap between B2 and B4 isolates the specific contribution of evolutionary search. The balanced seed genome already achieves high safety (0.900) but leaves substantial room for structural improvement (0.571 vs. 0.924). The GA discovers that high structure weight ($w_c \approx 0.74$–0.78) consistently produces better outcomes than the balanced initialisation, a non-obvious finding that would be difficult to arrive at through manual tuning alone. The finding also suggests that safety and empathy are more easily satisfied by the hard constraints and phrase bank design than by genome weight allocation, freeing the evolutionary search to concentrate on structural completeness.

These observations align with Hypothesis H5: hard safety constraints prevent the GA from exploiting structural shortcuts at the cost of safety, as evidenced by B4 maintaining safety parity with B2 (0.901 vs. 0.900) while dramatically improving structure.

5.6. *Ethical Risks and Deployment Considerations*

The deployment of optimisation-driven conversational agents in mental-health contexts raises ethical concerns that extend beyond algorithmic performance. This subsection critically examines four deployment risks specific to the proposed framework and the Bangladeshi context.

Proxy metric misalignment. The empathy and safety scores used in this work are rule-based proxies derived from keyword detection. A response can achieve a high empathy score by including the phrase “it makes sense” without conveying genuine emotional attunement, and can achieve a perfect safety score by including the word “professional” without providing actionable crisis guidance. Optimising against these proxies risks producing responses that are *metrically safe* but not *clinically safe*. The gap between proxy score and genuine therapeutic value is difficult to close without human evaluation [3]. Any deployment of this system should therefore include ongoing monitoring by qualified mental health professionals who can identify cases where high-scoring responses are clinically inappropriate.

Over-reliance and substitution risk. Systems like Riri are designed as supplementary tools, not replacements for professional care. However, in Bangladesh, where access to mental health professionals is severely limited [1], users experiencing acute distress may interact with an AI assistant as their primary or only source of support. The high escalation frequency observed in our optimised system (100% structural component usage) means that every response includes a referral to professional help—but if no such help is accessible, this referral may feel dismissive rather than supportive. Future work should design context-aware escalation that acknowledges service availability limitations.

Cultural and linguistic validity. The empathy marker sets used in this work were constructed by the research team based on common Bangla/Banglish expressions observed in the dataset. They have not been validated by clinical psychologists or community mental health workers familiar with Bangladeshi emotional expression norms. Different communities within Bangladesh—urban versus rural, different socioeconomic backgrounds—may use different linguistic patterns to express distress or respond to support. A marker set that works well for the PHQ-4 screening population in the training dataset may not generalise to other populations. The cultural specificity of empathetic language is particularly important given that the optimised genome's high structure weight ($w_c^\star = 0.778$) means structural components dominate the response, potentially at the cost of culturally-appropriate emotional tone [14].

Reward hacking in deployment. The failure modes documented in this work—question-only collapse and empathy marker stuffing—were discovered through offline trace analysis before deployment. In a live deployment, new failure modes may emerge from interactions with real users whose language patterns differ from the training data. If the system is periodically re-optimised using interaction logs, the GA may discover new shortcuts that exploit patterns in real user behaviour. Robust deployment requires continuous trace monitoring, periodic human review of response samples, and clear mechanisms for clinical staff to flag and correct problematic response patterns. The interpretable genome design facilitates this: because generator behaviour is controlled by a small set of named parameters, targeted corrections can be made without rerunning the full optimisation pipeline.

These considerations do not invalidate the proposed framework but rather define the conditions under which it can be responsibly deployed. The framework provides a technically sound and data-efficient baseline for mental-health conversational optimisation. Responsible deployment requires it to be embedded within a broader governance structure that includes clinical oversight, user safety monitoring, and culturally-informed evaluation protocols [4,5].

5.7. Limitations and Future Work

Several limitations of the present work warrant attention:

- **Interpretable proxy metrics by design.** The empathy and safety metrics use rule-based keyword detection rather than learned evaluators. This design choice prioritises interpretability and transparency: because each score component corresponds to a named linguistic marker, it is possible to diagnose exactly why a response received a given score and to correct the scoring function when failure modes are identified. This property was essential for the iterative scoring redesign documented in Section 5. Future work should validate these proxies against human clinical judgement and explore hybrid approaches combining rule-based interpretability with learned scoring accuracy.
- **Single-turn interactions only.** All experiments evaluate single-turn prompt-response pairs. Real mental-health conversations are multi-turn, where risk levels may escalate across turns and emotional context accumulates. The current genome parametrises a single response policy; extending to multi-turn would require a sequential decision framework where the GA optimises a turn-level policy conditioned on conversation history. The `memory_window` gene in the current genome was designed with this extension in mind and provides a natural starting point for future work.
- **Limited cultural validation.** Bangla emotional expression norms are only partially captured by the current marker sets. Future work should involve community mental health workers in Bangladesh to validate and extend the marker banks.
- **Human evaluation conducted.** A structured annotation study with five clinical psychologists (30 stratified response pairs, double-blind) confirms that the GA-optimised system substantially outperforms the zero-shot baseline on all three clinical dimensions ($p < 0.001$). Full results are reported in Section 4.13.
- **No personalisation.** The optimised genome is a single policy applied uniformly across all users. Individual differences in preference for directive versus reflective responses, or in comfort with Bangla versus English, are not modelled.

Future extensions include: (1) multi-turn sequential genome policies, where the GA optimises escalation and structure decisions conditioned on full conversation history—the most impactful near-term research direction; (2) multi-objective evolutionary algorithms maintaining Pareto fronts of empathy, safety, and structure to explicitly model trade-offs rather than collapsing them into a scalar fitness; (3) learned empathy evaluators trained on clinically validated datasets, replacing the current rule-based proxy scores; (4) human preference alignment via direct annotator feedback integrated into the fitness function; (5) clinician-in-the-loop evaluation with mental health professionals rating response quality in a live deployment setting; and (6) multilingual expansion beyond Bangla/Banglish/English to other South Asian languages where mental-health support infrastructure is similarly limited.

6. Conclusions

We presented a risk-aware genetic optimization framework for a mental-health conversational agent operating in Bangla, Banglish, and English. By evolving an interpretable genome controlling a structured generator, the system achieves substantial improvements in response structure and overall fitness while maintaining high safety and stable empathy. The study illustrates how evolutionary algorithms can optimize non-differentiable, value-laden objectives and highlights the critical role of scoring design, trace-based evaluation, and safety constraints in responsible deployment of AI systems for mental-health support. The framework's data efficiency, interpretability, and deployment feasibility on standard hardware make it a practical foundation for resource-constrained mental-health applications in multilingual settings.

Appendix A. Evolutionary Failure Mode Examples

This appendix provides concrete examples of the two evolutionary failure modes documented in Section 5: question-only structural collapse and empathy marker stuffing. These failure modes were discovered through trace analysis during early development, before the scoring function was redesigned. The examples below are representative of the response patterns produced by high-fitness genomes under the initial (flawed) scoring design.

A.1 Question-Only Structural Collapse

Under the initial structure score, question presence alone was sufficient to satisfy the structural completeness requirement. Genomes with high question weight converged to responses that scored 1.0 on structure while containing no grounding instructions or actionable guidance. A representative example:

> *User:* "I feel really anxious and overwhelmed. Family niye onek stress e achi. What should I do now?"
>
> *Failed response (pre-fix, high structure score):* "How long have you been feeling this way? What do you think is causing the most stress? Have you talked to anyone about this? Is there something specific that happened recently? What usually helps you feel better?"
>
> *Structure score:* 1.0 *Empathy score:* 0.05 *Safety score:* 0.60

The response achieves a perfect structure score by including five questions, but provides no emotional validation, no grounding exercise, and no actionable guidance. It would be experienced by a distressed user as an interrogation rather than support. This failure mode was corrected by introducing the question-only penalty term $\delta = 0.20$ (Equation 13), which deducts from the structure score when a question is present but grounding and action components are absent.

A.2 Empathy Marker Stuffing

Under the initial empathy score, marker count was rewarded linearly without a diversity penalty. Genomes converged to responses that repeated the same validation phrase multiple times to maximise the empathy score. A representative example:

> *User:* "Ami onek lonely feel kortesi lately. I feel really anxious and overwhelmed."
>
> *Failed response (pre-fix, high empathy score):* "It makes sense that you feel this way. It makes sense to feel overwhelmed. It makes sense that this is hard. It makes sense that you're struggling. What would help you right now?"
>
> *Empathy score:* 0.90 *Structure score:* 0.20

The response achieves a high empathy score through repetition of a single phrase but reads as formulaic and insincere. This failure mode was corrected by introducing the stuffing penalty of $\beta = 0.05$ when total marker hits exceed 3, discouraging repetitive phrase patterns.

A.3 Post-Fix Optimised Response (Same Prompt)

After scoring redesign, the same prompt from A.2 produces:

> *Optimised response (post-fix):* "Feeling this alone can be really painful. তুমি একা না। First, try to sit somewhere comfortable and take one slow breath. Try not to stay completely isolated with these thoughts. Is there someone nearby you feel comfortable talking to?"
>
> *Empathy score:* 0.45 *Structure score:* 1.0 *Safety score:* 1.0

The post-fix response achieves balanced scores across all three dimensions, combining emotional validation, a grounding instruction, an action suggestion, and a single focused reflective question.

Appendix B. Optimised Genome Evolution Examples

Table A1 shows representative genome configurations at three stages of the GA run: the balanced seed initialisation, an intermediate generation, and the final converged best genome.

Table A1. Representative genome configurations at three GA stages. w_e, w_s, w_c are empathy, safety, and structure weights (renormalised to sum to 1). θ_{mid}, θ_{high} are risk escalation thresholds.

Stage	w_e	w_s	w_c	θ_{mid}	θ_{high}	**Fitness**
Seed g_{bal} (Gen 0)	0.330	0.330	0.340	0.550	0.800	0.532
Intermediate (Gen 10)	0.152	0.100	0.748	0.697	0.700	0.678
Converged best (Gen 20)	0.137	0.086	0.778	0.700	0.731	0.678

The evolution shows a clear directional shift: w_c increases from 0.340 to 0.778 across 20 generations, while w_e and w_s decrease to their minimum floor values. This reflects the fitness landscape discussed in Section 3.8: safety is largely satisfied by the hard constraint and escalation logic, and empathy by the phrase bank design, leaving structural completeness as the primary differentiator between genomes. The GA consistently discovers this structure-dominant configuration across all five independent runs, indicating a stable global optimum rather than a run-specific artefact.

Appendix C. Dataset Composition

Table A2 summarises the composition of the full $N = 2297$ prompt dataset used in all experiments.

Table A2. Dataset composition ($N = 2297$ de-identified prompts). PHQ-4 severity bands: Normal (0–2), Mild (3–5), Moderate (6–8), Severe (9–12). Risk labels are derived from PHQ-4 total score and explicit self-harm indicator detection (Section 3).

Category	**Count**	**Proportion**
Risk label distribution		
High risk	1,516	46.2%
Mid risk	959	29.2%
Low risk	804	24.5%
PHQ-4 severity distribution		
Severe (9–12)	1,459	44.5%
Moderate (6–8)	996	30.4%
Mild (3–5)	555	16.9%
Normal (0–2)	269	8.2%
Language of interaction		
Bengali (Bangla)	3,107	94.8%†
English	137	4.2%
Banglish	33	1.0%
Age group		
18–22 years	1,186	36.2%
23–27 years	797	24.3%
28–34 years	485	14.8%
13–17 years	478	14.6%
35–44 years	217	6.6%
45+ years	116	3.5%
Gender		
Male	2,339	71.3%
Female	826	25.2%
Other	114	3.5%

† Language proportions are based on the full 3,279-row collection; 2,297 rows with valid prompts were used for experiments after filtering empty entries.

Several aspects of the dataset composition are noteworthy for interpreting the experimental results. First, the dataset is heavily skewed toward high-risk cases (46.2%), reflecting the clinical context of the application. Second, the dataset is predominantly young (74.5% aged 18–27), male (71.3%), and Bengali-speaking (94.8%), which limits generalisability of the empathy marker sets to other populations. Third, the high proportion of severe PHQ-4 scores (44.5% scoring 9–12) confirms that the dataset captures genuine clinical distress rather than mild wellbeing concerns.

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Institutional Review Board Statement: The dataset used in this study consists of de-identified prompts collected as part of routine mental-health screening and support services in Bangladesh. No personally identifiable information was retained. As the data were collected in the course of standard service delivery and fully anonymised prior to research use, formal ethical review was not required under applicable institutional guidelines. The study was conducted in accordance with the Declaration of Helsinki.

Informed Consent Statement: Not applicable. The dataset comprises fully de-identified text prompts with no personally identifiable information. No individual participants can be identified from the published data.

Data Availability Statement: The dataset used in this study is not publicly available due to patient privacy restrictions and the sensitive nature of mental-health screening data. The source code for the EvoRiri framework, including the genetic algorithm, response generator, and scoring functions, is available from the corresponding author upon reasonable request.

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References

1. Qiyang Zhang, Renwen Zhang, Yiying Xiong, Yuan Sui, Chang Tong, and Fu-Hung Lin. Generative AI mental health chatbots as therapeutic tools: Systematic review and meta-analysis of their role in reducing mental health issues. *Journal of Medical Internet Research*, 27:e78238, 2025.
2. Wilson Kin Chung Leung, Simon Ching Lam, Bobo Ching Lam Chan, Janice Ngar Lam Chow, and Yvonne Yuet Ying Wong. Chatbot interventions for improving mental health among people in Asia: A systematic review and meta-analysis of randomised controlled trials. *BMJ Health Care Informatics*, 33(1):e101479, 2026.
3. Ruvini Sanjeewa, Ravi Iyer, Pragalathan Apputhurai, Nilmini Wickramasinghe, and Denny Meyer. Empathic conversational agent platform designs and their evaluation in the context of mental health: Systematic review. *JMIR Mental Health*, 11:e58974, 2024.
4. American Psychological Association. Use of generative AI chatbots and wellness applications for mental health: APA health advisory. Available online: https://www.apa.org, 2024.
5. Julian De Freitas, Ahmet Kaan Uğuralp, Zeliha Oğuz-Uğuralp, and Stefano Puntoni. Chatbots and mental health: Insights into the safety of generative AI. *Journal of Consumer Psychology*, 34(3):481–491, 2024.
6. Yisong Zhang, Ran Cheng, Guoxing Yi, and Kay Chen Tan. A systematic survey on large language models for evolutionary optimization: From modeling to solving. *arXiv preprint arXiv:2509.08269*, 2025. Preprint; submitted for journal review.
7. Chao Wang, Jiaxuan Zhao, Licheng Jiao, Lingling Li, Fang Liu, and Shuyuan Yang. When large language models meet evolutionary algorithms: Potential enhancements and challenges. *Research*, 8:0646, 2025.
8. Qinglong Hu, Tong Xialiang, Mingxuan Yuan, Fei Liu, Zhichao Lu, and Qingfu Zhang. Multimodal LLM-assisted evolutionary search for programmatic control policies. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026.
9. Pia Chouayfati, Niklas Herbster, Ábel Domonkos Sáfrán, and Matthias Grabmair. GenDLN: Evolutionary algorithm-based stacked LLM framework for joint prompt optimization. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*

(Volume 4: Student Research Workshop), pages 1171–1212, Vienna, Austria, 2025. Association for Computational Linguistics.

10. Chrisantha Fernando, Dylan Banarse, Henryk Michalewski, Simon Osindero, and Tim Rocktäschel. Promptbreeder: Self-referential self-improvement via prompt evolution. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 13481–13544. PMLR, 2024.
11. Xavier Sécheresse, Jacques-Yves Guilbert-Ly, and Antoine Villedieu de Torcy. GAAPO: Genetic algorithmic applied to prompt optimization. *Frontiers in Artificial Intelligence*, 8:1613007, 2025.
12. Ruwini Herath. Emotionally intelligent chatbots in mental health: A review of psychological, ethical, and developmental impacts. *International Journal of Computer Applications*, 187(29):49–56, 2025.
13. Kathleen Kara Fitzpatrick, Alison Darcy, and Molly Vierhile. Delivering cognitive behavior therapy to young adults with symptoms of depression and anxiety using a fully automated conversational agent (Woebot): A randomized controlled trial. *JMIR Mental Health*, 4(2):e19, 2017.
14. R. AlMakinah, A. Norcini-Pala, L. Disney, and M. A. Canbaz. Enhancing mental health support through human-AI collaboration: Toward secure and empathetic AI-enabled chatbots. *arXiv preprint arXiv:2410.02783*, 2024. Preprint; submitted to ACM CHI.
15. Sharjeel Tahir, Shahid Mahmood Bhatti, and Sami Azam. Artificial empathy classification: A survey of deep learning techniques, datasets, and evaluation scales. *arXiv preprint arXiv:2310.00010*, 2023. Preprint.
16. Nastaran Saffaryazdi, Elnaz Gharavi, Niloofar Gharavi, Mark Billinghurst, and Suranga Nanayakkara. Empathetic conversational agents: Utilizing neural and physiological signals for enhanced empathetic interactions. *International Journal of Human–Computer Studies*, 198:103455, 2025.
17. Ibidun Christiana Afolabi and Olasunkanmi Tobi Afolabi. Conversational agents: An exploration into chatbot evolution, architecture, and important techniques. In *Proceedings of the International Conference on Technology (IConTech)*, 2024.
18. Thomas Bäck. *Evolutionary Algorithms in Theory and Practice.* Oxford University Press, 1997.
19. Mir Md. Jahangir Kabir, Shuxiang Xu, Byeong Ho Kang, and Zongyuan Zhao. A new multiple seeds based genetic algorithm for discovering a set of interesting Boolean association rules. *Expert Systems With Applications*, 74:55–69, 2017.
20. Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. mT5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 483–498, 2021.
21. Abhik Bhattacharjee, Tahmid Hasan, Wasi Uddin Ahmad, Kazi Samin Mubasshir, Md Saiful Islam, Anindya Iqbal, M. Sohel Rahman, and Rifat Shahriyar. BanglaNLG and BanglaT5: Benchmarks and resources for evaluating low-resource natural language generation in Bangla. In *Findings of the Association for Computational Linguistics: ACL 2023*, 2023.
22. Shahidul Islam. BanglaGPT: A GPT-2 based language model for Bangla, 2023.
23. Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945.
24. Klaus Krippendorff. Computing Krippendorff's alpha-reliability. *Departmental Papers (ASC), University of Pennsylvania*, 2011.